Utilization of Onboard Diagnostic II (OBD-II) on Four Wheel Vehicles for Car Data Recorder Prototype

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***Abstract*— The government has required public transport provider to equip their fleets with information tools if their fleets get in an accident. Currently, the collection of evidence of motor vehicle accidents by the authorities has not included data on the condition of the vehicle before the occurrence of the accident. Unlike an aircraft accident investigation that includes aircraft condition data derived from various sensors recorded in Flight Data Recorder (FDR). The recording data in FDR can be the evidence of the accident and it is expected to add to the accuracy of an accident investigation. Car Data Recorder Prototype is a system capable of recording the condition of four-wheeled vehicles and notify if there has been an accident. Vehicle condition recording utilizes Onboard Diagnostic-II (OBD-II) feature with parameter recording: gas pedal position, engine speed, vehicle speed, and engine temperature. Accident detection utilizes the ADXL345 accelerometer sensor and the condition of open or non-open airbag vehicles monitored by Diagnostic Trouble Code (DTC). If there is an indication of an accident, notification in the form of SMS will be sent by using GSM SIM800L module. Testing the data recording feature prototype car data recorder performed on city car is able to take advantage of features OBD-II to record the parameters that have been determined. Recorded data is processed into graphical information showing the position of the gas pedal, engine speed, vehicle speed, and engine temperature within a specified time range. There is a 5-second delay in recording data of engine speed and speed of the vehicle, but the data pattern of recording engine rotation is identical with 84.8% accuracy while the data of vehicle speed recording is also synonymous with 74.4% accuracy. Based on the accident simulation results, the ADXL345 and DTC accelerometer can detect frontal collisions and airbag conditions, which triggers GSM SIM800L to send SMS notification.**

Keywords*— *accident; investigation; OBD-II; accelerometer; airbag; DTC

I. INTRODUCTION

Traffic accidents are unexpected and accidental incidents on roads involving vehicles with or without other road users resulting in human casualties and/or property loss [1]. There are many factors that can cause traffic accidents such as undisciplined driving behavior, poor road environment, or due to inadequate vehicle condition resulting from poor maintenance.

The Government has required public transport providers to equip their fleets with an accident information tool written in Law of the Republic of Indonesia Number 22 of 2009 on Road Traffic and Transportation Article 204 paragraph (2) containing "General Motor Vehicles must be equipped with tools of Traffic Accident information to the Control Center for Traffic Safety and Road Transport Systems" [2].

According to the Regulation of the Chief of Police of the Republic of Indonesia Number 15 of 2013 on the Procedures for Handling Traffic Accidents Articles 26 and 27, the collection of evidence of motor vehicle accidents has not mentioned the condition of the vehicle before the accident occurred [3] so that the accuracy of investigation results is not maximal. In contrast to aircraft accident investigations that have included evidence of aircraft condition data derived from various sensors recorded by Flight Data Recorder (FDR) [4] so as to increase the accuracy of the investigation results.

In this modern era, motor vehicles, especially four-wheel vehicles have been equipped with the Engine Control Unit (ECU) and various kinds of electronic sensors, both the engine sector, transmission, and chassis. The data released by the sensor will be processed by the ECU to be the trigger of an actuator in the vehicle system [5]. Because modern motor vehicles have been equipped with various sensors and ECUs, motor vehicle manufacturer supplements their products with Onboard Diagnostic II (OBD-II) [5] [6]. OBD-II is used for monitor vehicle condition to diagnose damage to the vehicle by reading Diagnostic Trouble Code (DTC) [6].

The CDR prototype will utilize sensor and DTC data obtained from OBD-II. The accelerometer will be used to detect accidents. The CDR prototype is expected to help investigate accidents and detect vehicle accidents.

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II. THEORY

A. Embedded System

Embedded System is a combination of hardware and software designed with specific functions for larger systems such as industrial machines, motor vehicles, medical equipment, cameras, and more. Designing the embedded systems requires the size, power consumption, system reliability, and performance because it is designed only for specific functions [7].

Embedded systems have a basic property such as computer namely receiving inputs, processing inputs, and producing output. In general, the input on embedded systems in the form of certain sensors depends on the needs of the system. The input will then be processed microcontroller/microprocessor which is already programmed for a specific function. The output of the embedded system can be an actuator or just an indicator. Embedded systems can function with little or no contact with humans [7].

Programming embedded systems using computers to write programs. Programming languages on embedded systems generally use multiple languages, such as Assembly, C, or Python by using the compiler according to the platform used.

B. OBD-II

In the early 1980s, four-wheeled automobile manufacturers started to implement various electronic devices in their vehicles. The main objective is to reduce exhaust emissions and increase fuel efficiency. To achieve these objectives, the system constantly monitors various sensors that are related to exhaust emissions and fuel efficiency. The electronic devices also have a function to help diagnose the damage to the vehicle engine. When damage is detected, the system must be able to store details of the damage which will then be analyzed by the technician and turn on the machine malfunction indicator or Malfunction Indicator Light (MIL) on the dashboard of the vehicle.

Initially, each manufacturer has its own electronic diagnostic standard. This makes the consumer depend on the vehicle manufacturer in case of damage to the vehicle because only the manufacturer's workshop can access the vehicle's electronic system information. Because of this problem, a standard communication protocol between the ECU and the vehicle diagnostic tool is required. OBD-II is the answer to the problem.

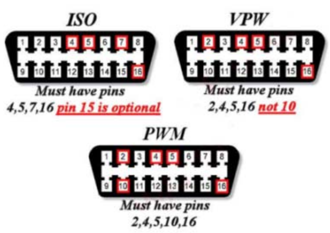

Fig. 1. Arrangement of pin on various OBD-II protocols

There are five protocols that can be used to communicate with the ECU. The protocols can be implemented on the vehicles, namely [5] [9] [10]:

- SAE J1850 PWM
- SAE J1850 VPW
- ISO 9141
- ISO 14230 KWP2000
- ISO 15765 (CAN)

It should be noted that one type of vehicle only implements one protocol. The most significant difference of the five protocols above is the placement of pins for communication [5] [9] as seen in Figure 1.

III. SYSTEM DESIGN

A. Hardware Design

Prototype Car Data Recorder uses ELM327 Bluetooth OBD-II Scanner, Arduino MEGA 2560, Bluetooth HC-05, ADXL345 Accelerometer, Datalogger + RTC, and GSM Module SIM800L. The entire hardware is placed on two 4x2.1 inch circuit board that is mounted on Arduino Mega 2560. The Figure 2 is a block diagram design of the prototype hardware:

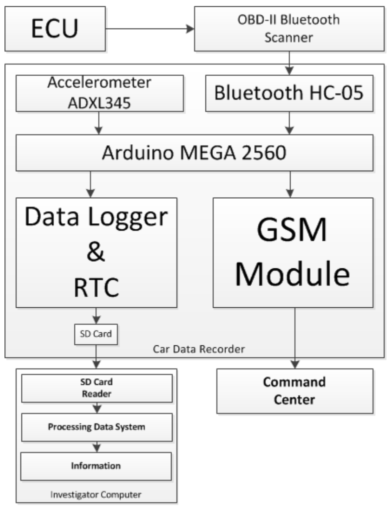

Fig. 2. Block diagram of CDR hardware design

ELM327 Bluetooth OBD-II Scanner serves to translate OBD-II data format on vehicle ECU into hexadecimal digits. To be able to communicate with Arduino Mega 2560, Bluetooth HC-05 module is used. The ADXL345 accelerometer is used to detect frontal collision that indicates accidents based on the G-force parameter. OBD-II data is time-stamped by Real Time Clock (RTC) and stored in SD Card by the data logger. If an accident is detected by the system, then the notification in the form of SMS will be sent to a number that has been determined through GSM module.

On the shield CDR comm circuit board there is a micro USB socket for power input, GSM SIM800L module, GSM antenna, and voltage regulator. A voltage regulator is required because the GSM SIM800L module works on a voltage range of 3.4 volts - 4.2 volts DC, while the system uses a 5 volt DC voltage. The current regulator is used to providing current up to 3 amperes. There is an available place for GPS and external

GPS antenna for further prototype development on this circuit board as seen in Figure 3.

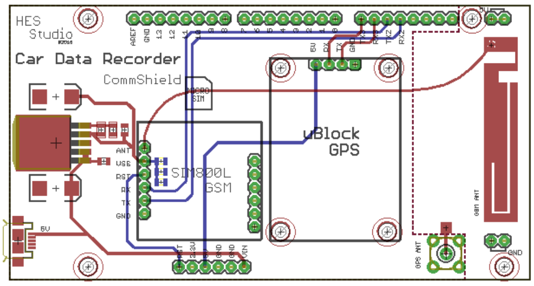

Fig. 3. Circuit Board CDR Comm Shield

On the circuit board car data recorder system shield (see Figure 4), there are three hardware, namely Bluetooth HC-05, data logger, and accelerometer ADXL345. In this circuit board, there is also a transistor to amplify the interrupt pin current on the ADXL345 accelerometer. All of the hardware on this circuit board work on a voltage of 5 VDC.

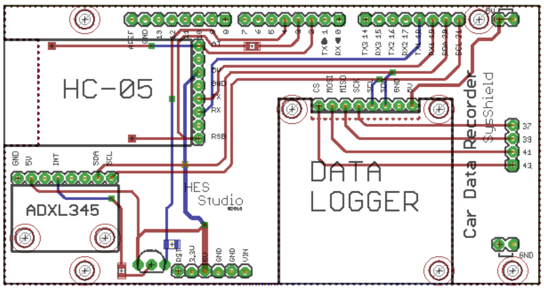

Fig. 4. Sys Shield CDR circuit board

B. *Software Design*

The design of CDR prototype software is divided into three functions (see Figure 5), which are able to communicate with the vehicle ECU, able to detect accidents and able to record OBD-II vehicle data.

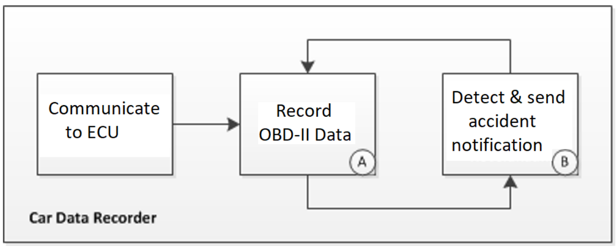

Fig. 5. CDR prototype system flowchart

Based on Figure 6, the process begins by searching the OBD-II Scanner using Bluetooth. In order to communicate with the ECU, the system will automatically search for the OBD-II protocol used on the vehicle. If the vehicle does not use all protocols according to OBD-II standard supported by OBD-II Scanner then the system will provide incompatible messages. If supported, the system will wait for the engine to turn on with the engine spin indicator (RPM>0). If the machine is on, the CDR system starts.

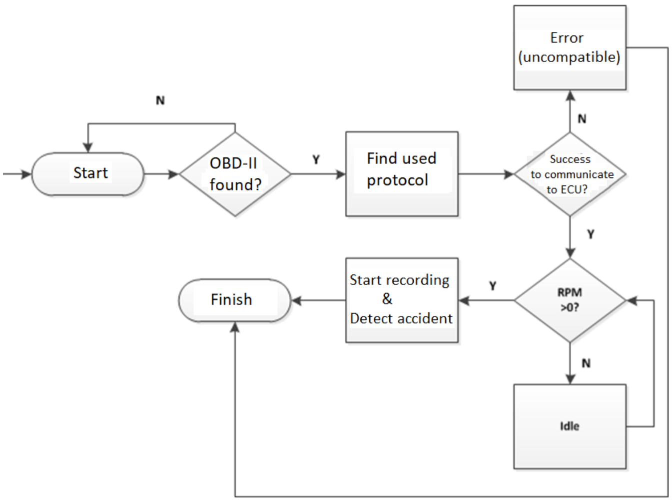

Fig. 6. Connection between ECU & CDR prototype flowchart

Based on Figure 7, Diagnostic Trouble Code (DTC) parameter is used for accident detection, while airbag and G-Force condition from the Accelerometer ADXL345. If an accident is detected (active airbag or G-force>4G), the accident notification will be sent to a pre-determined number via SMS Gateway GSM Module.

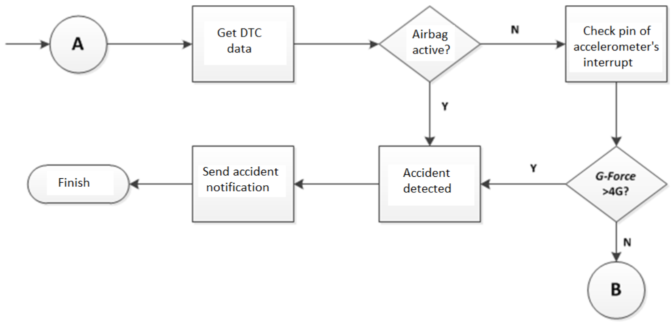

Fig. 7. Accident detection & sending accident notification flowchart

Based on Figure 8, the process begins with OBD-II data retrieval. The received data is a hexadecimal digit that must be processed according to the ELM327 OBD-II Scanner datasheet. The data has been processed and then given a time stamp and then stored.

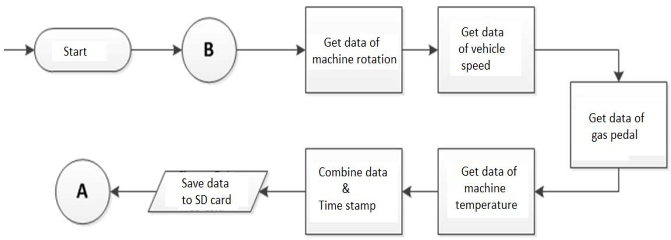

Fig. 8. Data Record Flowchart

IV. RESULT

A. *Testing of Reading Gas Pedal Positioning, Machine Rotation, and Speed*

The validation testing of the position of the accelerator is carried out with the aid of measured depth of the gas pedal depth for 0% to 100%. The position of the gas pedal of the OBD-II is 2 hexadecimal digits then converted to percent unit (%) by the formula [11]:

$$Trottle\ Position = \frac{100}{255}A \tag{1}$$

A = Result of conversion of hexadecimal number to decimal

Here is the result of testing the gas pedal position validation of data from OBD-II.

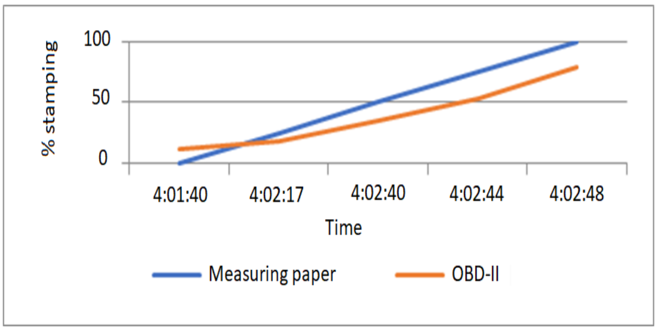

Fig. 9. Graph of gas pedal position testing

On the graph (Figure 9), there is a difference when the gas pedal is not stepped on (0%) because the position of the throttle body on the vehicle is slightly open for the required air intake of the engine when stationary. When the gas pedal stepped on the position of 25% to 100%, it can be seen that the pattern of both data is identical.

The testing of machine rotation and the engine speed is done by matching the data between the video (camera) machine rotation and vehicle speed when testing with OBD-II data. Machine rotation data from OBD-II is 4 hexadecimal digits, while the speed data is 2 hexadecimal digits. To convert these hexadecimal data into Rotation per Minute (RPM) and Km/h units are done by the formula [11]:

$$Engine\ RPM = {}^{A}/_{4} \tag{2}$$

A = Result of conversion of 4 hexadecimal numbers to decimal

$$Vehicle\ Speed = B \tag{3}$$

B = Result of conversion of 2 hexadecimal numbers to decimal

Here is a comparison of the results of the engine spin testing:

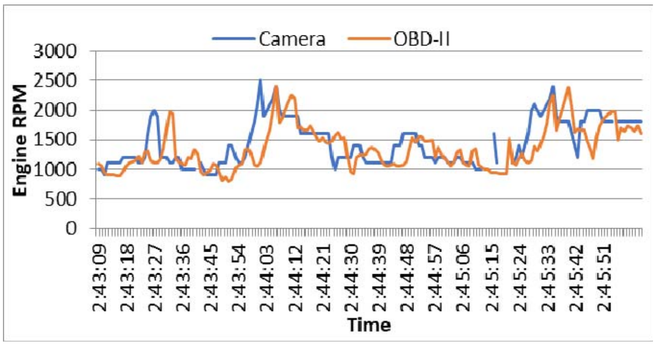

Fig. 10. Graph of machine rotation testing

Here is a comparison of vehicle speed test results:

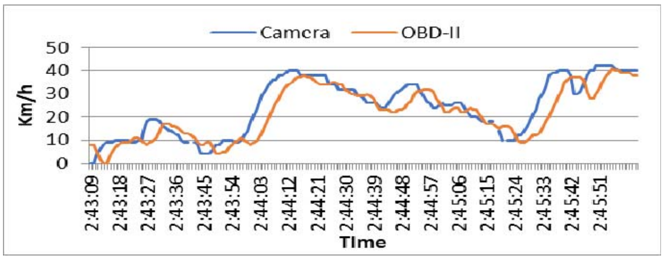

Fig. 11. Graph of vehicle speed testing

From the test result, the data pattern of CDR prototype recording is identical with actual condition and has accuracy for engine speed equal to 84.8% (see Figure 10) and speed equal to 74.4% (see Figure 11). Accuracy rate is affected by the human error in the form of camera position during recording and tachometer needle reading and speedometer. From the graph, there is a delay of 5 seconds between the actual conditions with the recording. Delay is caused by several things that the timestamp is given is the last data timestamp (engine temperature), response time ELM237 Bluetooth OBD-II Scanner of 10-59 ms is still less fast, and the performance of Arduino Mega 2560 which is only 8 bits with a speed of 16 MHz.

The engine temperature test is done by comparing the temperature read by the DS18B20 temperature sensor placed on the vehicle radiator. Data from the DS18B20 sensor is stored in micro SD by using Arduino ProMini microcontroller. Period of taking data per one second for five minutes. The temperature reading through OBD-II produces 2 hexadecimal digits. To obtain a measurement result with a degree of degrees Celsius (°C), then the conversion is done by the formula [11]:

$$Engine\ Temperature = A - 40 \tag{4}$$

A = Conversion result 2 digit hexadecimal number to decimal

Based on the test results (Figure 12), there is a difference in temperature reading ± 30 °C, this is because the temperature sensor DS18B20 comparator placed on the inlet hose radiator made of rubber and temperature sensors DS18B20 not directly exposed to direct radiator water. The difference in the graphic pattern at the beginning (0^{th} data) is caused because the water of the radiator read by OBD-II has not been subjected to radiator cooling, while the DS18B20 comparator temperature sensor has been exposed to cooling by the radiator. After the machine's working temperature stabilizes, the graph pattern of the two data is relatively identical with the difference of ± 30 °C data.

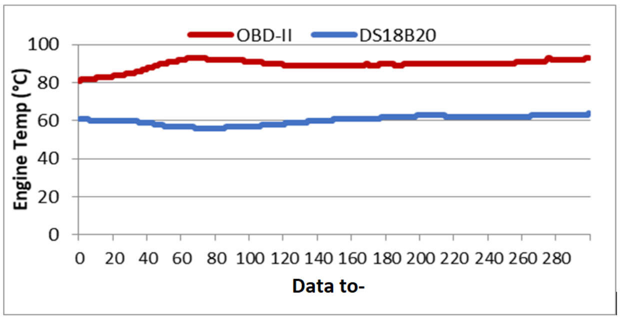

Fig. 12. Graph of machine temperature testing

B. *Accident Detection Testing*

The accelerometer function testing is done by putting the prototype in a safe pack, then being thrown horizontally to the wall from distance of one, three, and five meters. This test scenario aims to validate the ADXL345 accelerometer function and SMS notification that simulates accidental events. The following test results have been done:

TABLE I. ACCELEROMETER TEST RESULTS AND SMS NOTIFICATION

Distance	SMS Sent
1 Meter	✓
3 Meters	✓
5 Meters	✓

Based on the Table I., it is proven that the accelerometer can detect any collisions (accidents), thus the SMS notifications can be sent.

DTC Testing Airbag is done with the help of Freematics OBD-II Emulator MK2 which can simulate DTC state on four-wheeled vehicles. When an accident occurs OBD-II will generate the code B1785 for the left airbag and B1786 for the right airbag. Here are the results of DTC airbag reading test:

TABLE II. DTC AIRBAG AND SMS NOTIFICATION TESTING RESULTS

Position	DTC	SMS Sent
Left	B1785	✓
Right	B1786	✓

Based on the Table II, it is proven that DTC Airbag can detect any collision (accident) so SMS notification can be sent.

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